Statistics 651
Posterior Approximation

Winter 2026

BYU Stats

Objectives

1. Use asymptotic theory for normal approximation to the posterior distribution
2. Review tools for Bayesian computation and approximations to posterior quantities (alternatives to MCMC)

Resources

- BDA3:
  - Ch 4.1, 4.2 (normal approximation)
  - Ch 10.1-10.5 (basic computation)
  - Ch 13 (advanced modal and distributional approximations)

Normal approximation

Normal approximation

Idea:

- Posterior distributions can be approximated by normal distribution.
- Normal posterior is justified with a large sample size even when the likelihood is combined with a non-normal prior.
- **Bernstein–von Mises theorem** states that under certain conditions, a posterior distribution converges in total variation distance to a multivariate normal distribution centered at the maximum likelihood estimator with covariance matrix given by the Fisher information matrix evaluated at the true population parameter value (see Section 10.2 in van der Vaart, 1998).

Normal approximation

Motivation

Consider the log of the normal density

$$\log(f(y)) = -\frac{1}{2\sigma^2}(y-\mu)^2 + \text{const.}$$

What is the mode?

μ

What is $-\left[\frac{\mathrm{d}^2}{\mathrm{d}y^2}\log(f(y))\right]^{-1}$?

$$\frac{d}{dy}\left[-\frac{1}{2\sigma^2}\left(y^2 - 2y\mu + \mu^2\right) + \text{const}\right]$$

$$= -\frac{2y}{2\sigma^2} + \frac{2\mu}{2\sigma^2}$$

$$\frac{d^2}{dy^2} = -\frac{1}{\sigma^2} \longrightarrow -\left[\frac{d^2}{dy^2}\log(f(y))\right]^{-1} = \sigma^2$$

Normal approximation

Setup

Next, consider a Taylor series expansion of the log-posterior density. Assume the mode, $\hat{\theta}_{\text{MAP}}$ is in the interior of the support.

$\tilde{\theta}$

Use shorthand $\log(\pi(\theta \mid \boldsymbol{x})) := l_\pi(\theta)$.

$$l_\pi(\theta) = \underbrace{l_\pi(\hat{\theta})}_{\text{const}} + \underbrace{(\theta - \hat{\theta})\left[\frac{d}{d\theta} l_\pi(\theta)\Big|_{\hat{\theta}}\right]}_{=0} + \boxed{\frac{1}{2}(\theta - \hat{\theta})^2\left[\frac{d^2}{d\theta^2} l_\pi(\theta)\Big|_{\hat{\theta}}\right]} + \dots$$

normal !

mean $\hat{\theta}$

var $-\left[\frac{d^2}{d\theta^2} l_\pi(\theta)\Big|_{\hat{\theta}}\right]^{-1}$

Normal approximation

Multivariate case (consequence of the Bernstein–von Mises theorem)

Let $\pi(\boldsymbol{\theta} \mid \boldsymbol{x})$ satisfy regularity conditions, including being continuous in $\boldsymbol{\theta}$ and having interior mode $\hat{\boldsymbol{\theta}}_{\text{MAP}} \in \mathbb{R}^D$.

As the sample size increases, $\pi(\boldsymbol{\theta} \mid \boldsymbol{x})$ approaches a multivariate normal distribution, that is,

$$\pi(\boldsymbol{\theta} \mid \boldsymbol{x}) \approx \text{Normal}_D(\boldsymbol{\theta};\, \hat{\boldsymbol{\theta}}_{\text{MAP}}\,, \hat{\boldsymbol{\Sigma}})\,,$$

where $\hat{\boldsymbol{\Sigma}} = \left[-\nabla^2 \log(\pi(\boldsymbol{\theta} \mid \boldsymbol{x}))\big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_{\text{MAP}}} \right]^{-1}.$

See also Laplace's approximation (Wikipedia).

For a more on Mernstein-von Mises theorem, see:

- Appendix B of BDA3
- Wikipedia page
- Section 10.2 in van der Vaart, 1998

Normal approximation

Also, consider the log posterior density under iid sampling $x_i \overset{\text{iid}}{\sim} F$:

$$\log(\pi(\theta \mid \boldsymbol{x})) = \log(\pi(\theta)) + \sum_{i=1}^{n} \log(f(x_i \mid \theta)) + \text{const.}$$

What happens as n becomes large?

$\log \pi(\theta \mid \underset{\sim}{x})$ is dominated by likelihood

Normal approximation

Considerations

- Asymptotic connection with classical approximations
- Fast approximation to the posterior distribution
- Normal is symmetric $\rightarrow$ may be inappropriate in small samples

Normal approximation

Example

Approximate the joint posterior distribution of (α, β) in the bioassay example with a bivariate normal distribution.

Evaluate the quality of the approximation.

Bayes computation

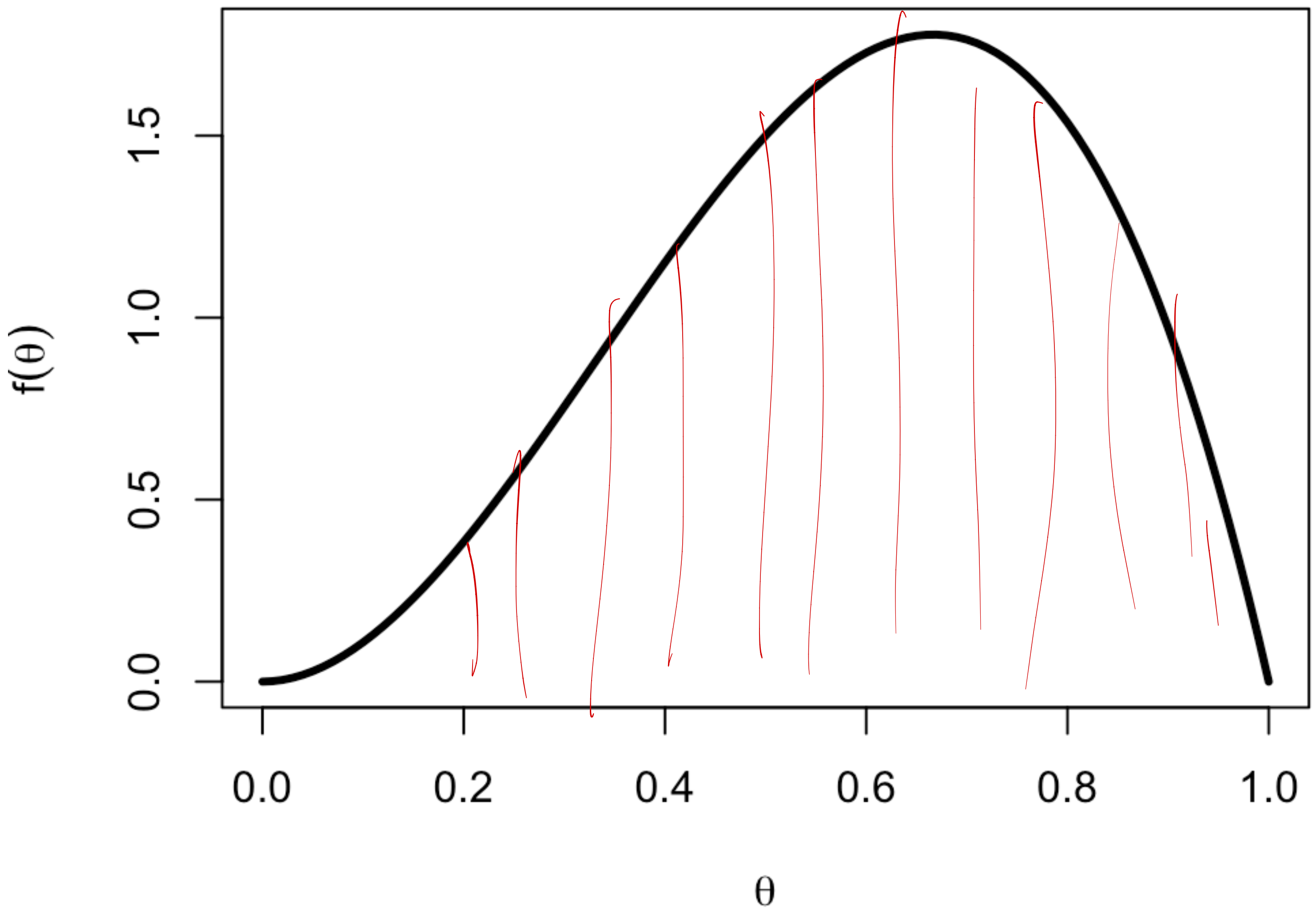
A note on notation

In this and future sections on computation, we seek to summarize or sample from what we call **target** distributions with densities $\pi(\boldsymbol{\theta})$.

These could refer to any probability distribution of interest, but in Bayesian inference, they typically refer to posterior distributions (we just suppress $\boldsymbol{\theta} \mid \boldsymbol{x}$ for brevity).

Numerical integration

Riemann integral



Numerical integration

Example

Consider density $\pi(\theta) \propto \theta^2(1-\theta)\,1\{0 < \theta < 1\}$.

Use numerical integration to

1. Estimate the normalizing constant.
2. Estimate $\mathbb{P}(\theta < 0.5)$.
3. Estimate $\mathbb{E}(\theta)$.

Sampling from distributions

Common families are implemented in most languages

Wonderful resource on sampling from distributions: Non-Uniform Random Variate Generation by Luc Devroye (1996)

Other common methods:

- Inverse CDF sampling
- Rejection sampling
- Importance resampling

Inverse CDF sampling

Proposition (Probability Integral Transform)

If F^{-1} is the inverse CDF of distribution F, then a draw

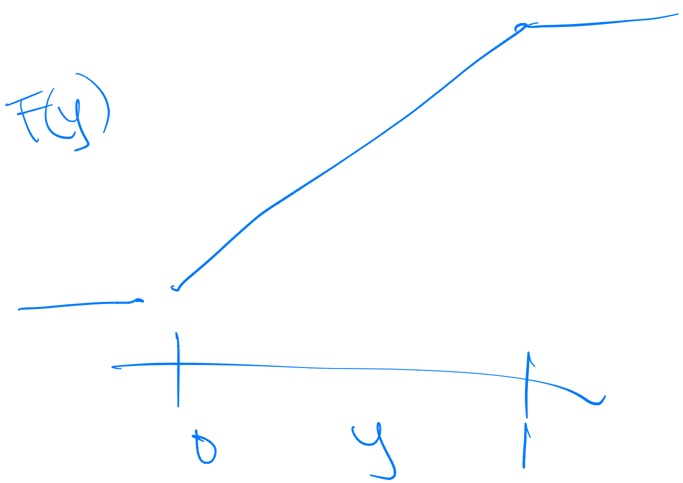
$\theta = F^{-1}(U)$ with $U \sim \mathrm{Uniform}(0, 1)$

is distributed according to F.

Proof:

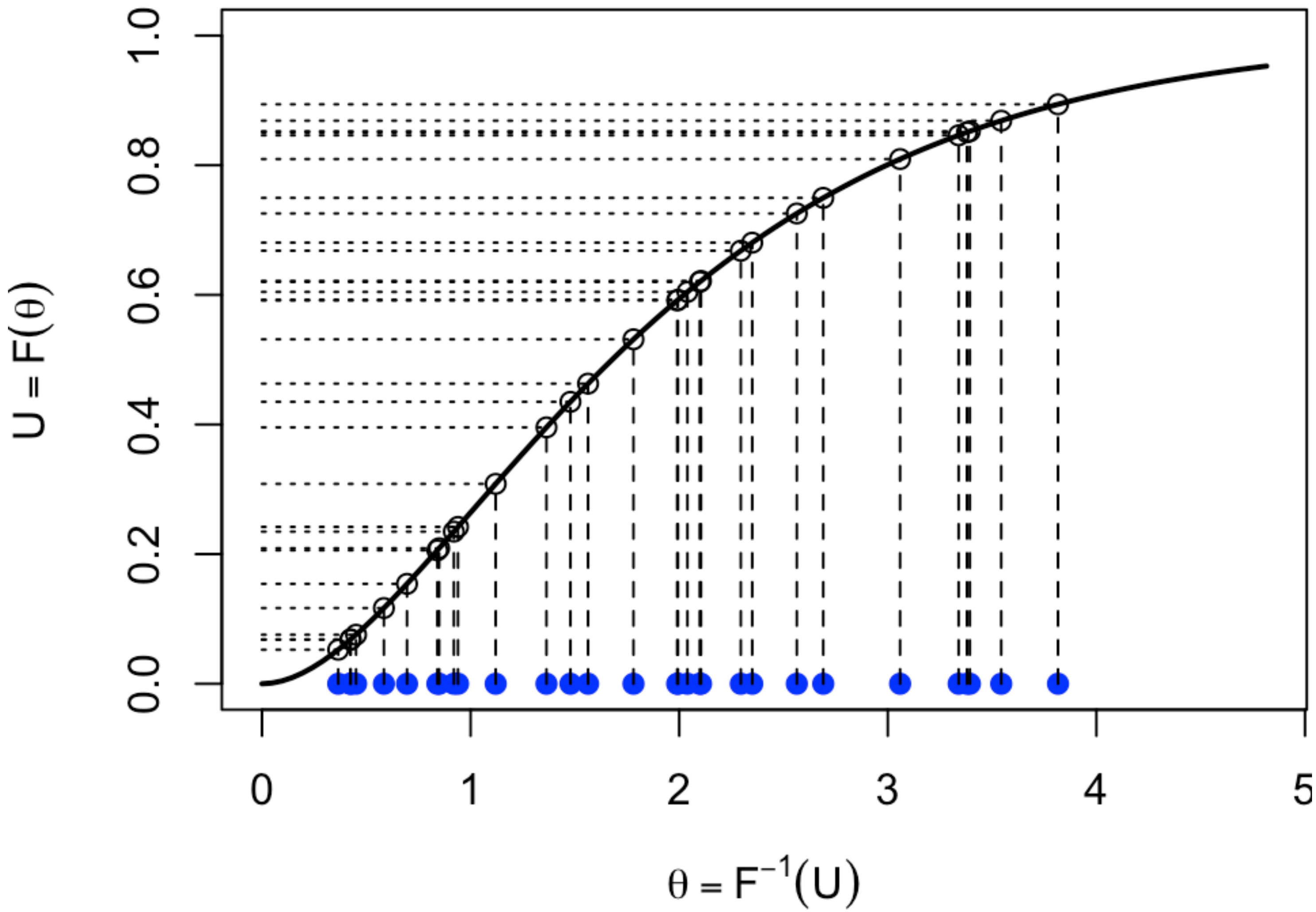
$$
\begin{aligned}
F_u(u) &= P(U < y) \\
&= P(F(\theta) < y) \\
&= P(\theta < F^{-1}(y)) \\
&= F(F^{-1}(y)) = y
\end{aligned}
$$

CDF of uniform



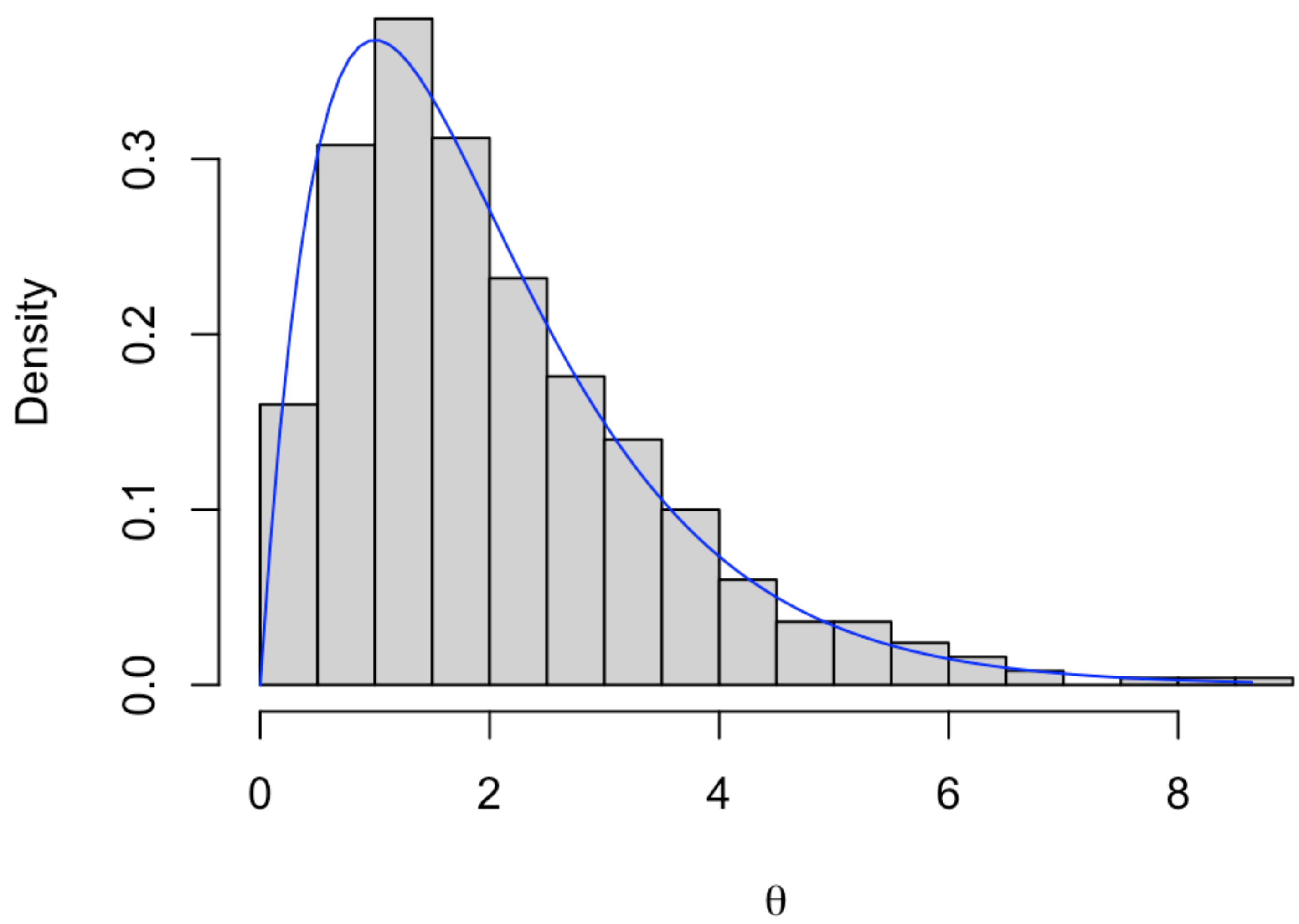
Inverse CDF sampling

▶ Code



Inverse CDF sampling

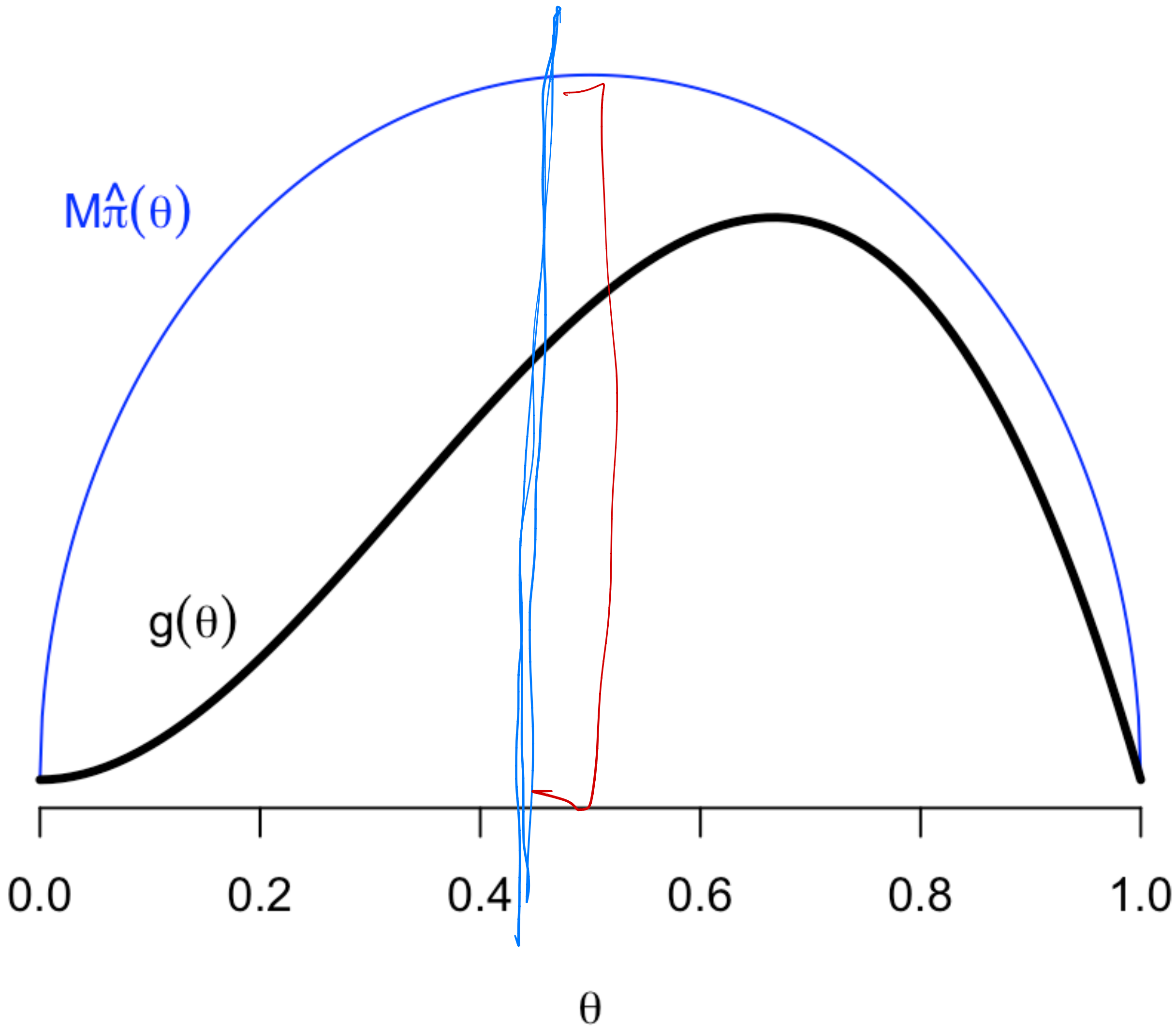
▶ Code



Rejection sampling

We have $g(\theta) \propto \pi(\theta)$.

User specifies $\hat{\pi}(\theta)$ that we can easily sample (called the **instrumental distribution**).



Rejection sampling

Algorithm

0. Find M such that $g(\theta) \leq M\,\hat{\pi}(\theta)$ for all θ.

Repeat:

1. Draw θ^* from $\hat{\pi}(\theta)$.
2. Draw $U \sim \mathrm{Unif}(0, M\hat{\pi}(\theta^*))$.
3. If $U < g(\theta^*)$, accept θ^*. Otherwise, reject.

The accepted draws $\{\theta^{(s)} : s = 1, \ldots, S\}$ come from the distribution proportional to $g(\theta)$.

Importance sampling

Suppose we have $c\,g(\theta) = \cancel{c}\,\pi(\theta)$, but c is unknown.

Introduce an instrumental distribution with density $\hat{\pi}(\theta)$, that

- approximates $\pi(\theta)$, and
- from which we can easily sample.

Then,

$$1 = \int c\, g(\theta)\, \mathrm{d}\theta = \int c \underbrace{\frac{g(\theta)}{\hat{\pi}(\theta)}}_{w(\theta)} \hat{\pi}(\theta)\, d\theta = c \int w(\theta)\, \hat{\pi}(\theta)\, d\theta$$

$$= c\, \mathbb{E}_{\hat{\pi}}\left[w(\theta) \right]$$

$$= c\, \frac{1}{S} \sum_{s=1}^{S} w\left(\theta^{(s)}\right)$$

$$\theta^{(s)} \overset{iid}{\sim} \hat{\pi}$$

Importance sampling

With an estimate $\hat{c} = \left[\frac{1}{S}\sum_{s=1}^{S} w(\theta^{(s)})\right]^{-1}$,

suppose we want to calculate $\mathbb{E}[h(\theta)]$.

$$\mathbb{E}[h(\theta)] = \int h(\theta)\, c\, g(\theta)\, \mathrm{d}\theta = c \int h(\theta) \underbrace{\frac{g(\theta)}{\hat{\pi}(\theta)}}_{w(\theta)} \hat{\pi}(\theta)\, d\theta$$

$$= c\, \mathbb{E}_{\hat{\pi}}\left[h(\theta)\, w(\theta)\right]$$

$$\approx \hat{c}\, \frac{1}{S} \sum_{s=1}^{S} h(\theta^{(s)})\, w(\theta^{(s)})$$

$$= \frac{1}{\frac{1}{S}\sum_{s} w(\theta^{(s)})} \frac{1}{S} \sum h(\theta^{(s)})\, w(\theta^{(s)})$$

$$= \sum \frac{w(\theta^{(s)})}{\sum_{s} w(\theta^{(s)})} h(\theta^{(s)})$$

Importance sampling

Algorithm

For $s = 1, \ldots, S$:

1. Draw $\theta^{(s)}$ from distribution with density $\hat{\pi}(\cdot)$
2. Calculate $w(\theta^{(s)}) := g(\theta^{(s)})/\hat{\pi}(\theta^{(s)})$

End loop

3. Normalize the weights $\tilde{w}_s = \frac{w(\theta^{(s)})}{\sum_{j=1}^{S} w(\theta^{(j)})}$

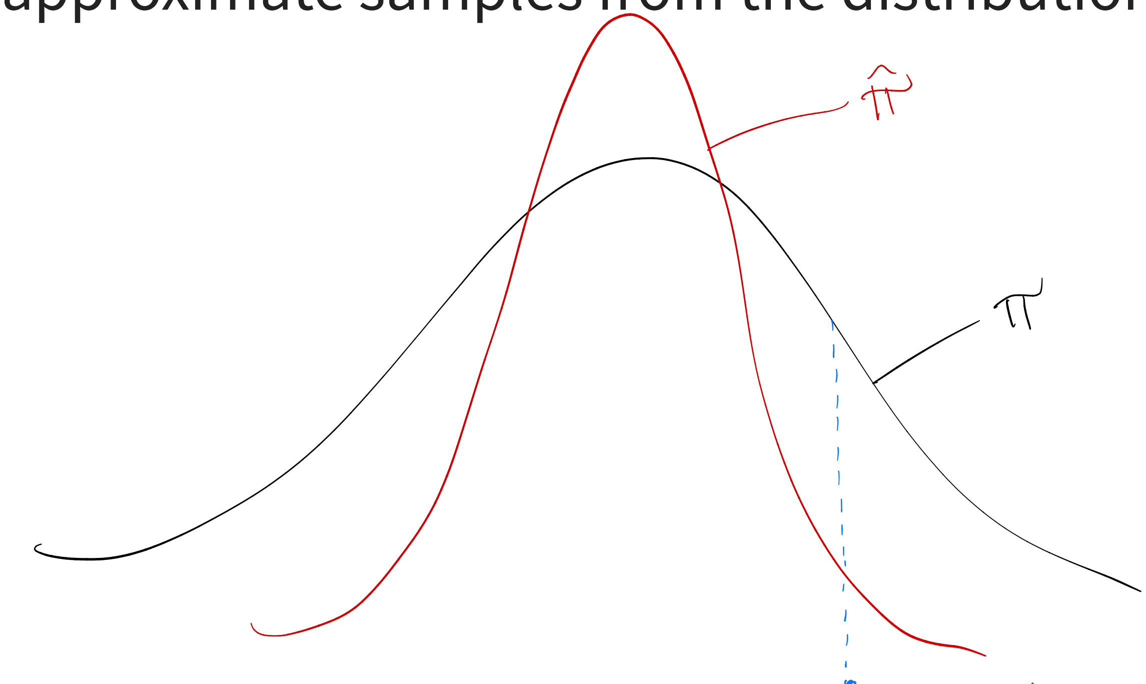
Output:

Estimate $\mathbb{E}[h(\theta)] \approx \sum_{s=1}^{S} \tilde{w}_s \, h(\theta^{(s)})$

Importance sampling

Notes

- Does not produce samples, but evaluates integrals
- Tails of $\hat{\pi}(\cdot)$ must be heavier than $\pi(\cdot)$.
- Great in one dimension. *How might it scale?*
- Could we get approximate samples from the distribution with density $\pi(\cdot)$?



occurs rarely and has large weight → bad...

Importance resampling

To get draws from the distribution having density $\pi(\cdot)\ldots$

Algorithm

1. Draw samples $\{\theta^{(s)} : s = 1, \ldots, S\}$ from distribution with density $\hat{\pi}(\cdot)$ and calculate weights $\{w(\theta^{(s)})\}$
2. Normalize the weights $\tilde{w}_s = \frac{w(\theta^{(s)})}{\sum_{j=1}^S w(\theta^{(j)})}$
3. Sample *without replacement* from a discrete distribution over $\{\theta^{(s)}\}$ with respective probabilities $\{\tilde{w}_s\}$.

Note: This requires that S be much larger than the size of the final sample.

Importance sampling

Example

Consider density $\pi(\theta) \propto \theta^2(1-\theta)\,1\{0<\theta<1\}$.

Use importance sampling to

1. Estimate the normalizing constant.
2. Estimate $\mathbb{P}(\theta < 0.5)$.
3. Estimate $\mathbb{E}(\theta)$.
4. Generate samples from $\pi(\theta)$ (with importance resampling).

See R code

Advanced approximation methods

- Variational inference (Ch 13.7 in BDA3)
  - Minimize Kullback-Leibler divergence between posterior and an easily specified distribution
  - Very fast
  - Good for point estimates, bad for uncertainty
- Approximate Bayesian computation (Ch 13.9 in BDA3)
  - Doesn't require a likelihood evaluation!
- Optimal transport (Li et al., 2025)
- Markov chain Monte Carlo (coming up!)